HelixQA Bank Run — helix_ota — Evidence Ledger

Revision: 1 **Last modified:** 2026-06-23T15:45:00Z

Phase-2 HelixQA autonomous-session run of `tools/helixqa/banks/helix_ota.yaml` (rev 6) against the LIVE system (§11.4.27). 20 scored challenges. Real captured evidence, honest §11.4.6 verdicts, §11.4.10 token redaction, §11.4.14 cleanup. NOT committed by this subagent (conductor commits).

Run context

- HEAD at run start: `1f6ec790`
- Live host (thinker live-system stack — owned by this run): `milosvasic@thinker.local` (x86_64, rootless podman 4.9.3 + podman-compose 1.0.6, §11.4.161).
- Shared in-memory ota-server booted once on `127.0.0.1:18091` for the 6 operational `<pw>` challenges (port 8080 was held by a sibling `htCore` process — left untouched).
- thinker F-CLUSTER (real ota-server + real PostgreSQL) self-booted per the 3 live-system challenges, run SERIALLY (single-resource-owner, §11.4.119 — the chaos challenge kills postgres, so no shared stack).

Evidence ledger (challenge | verdict | evidence path)

#	Challenge	Verdict	Evidence
1	HOTA-AUTH-LOGIN	PASS	docs/qa/20260623-helixqa-bank-run/operational-shared-server.txt (live :18091; 39/0/1)
2	HOTA-DEVICE-REGISTER	PASS	operational-shared-server.txt
3	HOTA-GROUP-LIFECYCLE	PASS	operational-shared-server.txt
4	HOTA-AUDIT-TRAIL	PASS	operational-shared-server.txt
5	HOTA-TELEMETRY-OVERVIEW	PASS	operational-shared-server.txt
6	HOTA-ROLLOUT-ROUTE-GATES	PASS	operational-shared-server.txt (1 expected pipeline SKIP-with-reason)
7		PASS	

#	Challenge	Verdict	Evidence
	HOTA-PIPELINE-SIGNED		HOTA-PIPELINE-SIGNED.log (32/0/0; free-port) + tests/e2e/PIPELINE_EVIDENCE.txt
8	HOTA-RECALL-LIFECYCLE	PASS	HOTA-RECALL-LIFECYCLE.log (35/0/0) + tests/e2e/RECALL_EVIDENCE.txt
9	HOTA-SECURITY-PROBES	PASS	HOTA-SECURITY-PROBES.log (37/0/0; free-port) + tests/security/RUN_EVIDENCE.txt
10	HOTA-SECURITY-PROBES-EXTENDED	PASS	HOTA-SECURITY-PROBES-EXTENDED.log (26/0/0) + tests/security/RUN_EVIDENCE_EXTENDED.txt
11	HOTA-FILTERS-PAGINATION	PASS	HOTA-FILTERS-PAGINATION.log (50/0/0) + tests/e2e/FILTERS_PAGINATION_EVIDENCE.txt
12	HOTA-AB-VIRT-BOOT	FAIL (host: QEMU+HVF accelerator non-functional — not a helix defect)	HOTA-AB-VIRT-BOOT.log; committed ref boot docs/qa/20260611T061626Z-ab-virt-boot/console.log
13	HOTA-AB-SLOT-SWITCH	FAIL (same host QEMU+HVF condition)	HOTA-AB-SLOT-SWITCH.log; committed ref docs/qa/20260611T094958Z-ab-slot-switch/verdict.txt
14	HOTA-AB-ROLLBACK	FAIL (same host QEMU+HVF condition)	HOTA-AB-ROLLBACK.log; committed ref docs/qa/20260611T095918Z-ab-rollback/verdict.txt
15	HOTA-RK3588-CONTROLPLANE	SKIP (exit 3, off-topology — board not adb-reachable; §11.4.52)	HOTA-RK3588-CONTROLPLANE.log; committed ref docs/qa/20260622-rk3588-controlplane/raw_validation_transcript.txt
16	HOTA-TRUST-BOUNDARY-LIVE	PASS	HOTA-TRUST-BOUNDARY-LIVE.log (4/0/0) + docs/qa/20260623-trust-boundary-live/SUMMARY.txt
17	HOTA-HTTP-LOAD-LIVE	PASS	HOTA-HTTP-LOAD-LIVE.log (4/0/0; p50=2.38ms p99=5.86ms non_2xx=0) + docs/qa/20260623-http-load-live/SUMMARY.txt
18	HOTA-CHAOS-LIVE	PASS	HOTA-CHAOS-LIVE.log (4/0/0; pg-kill+reconnect, corruption, contention, churn) + docs/qa/20260623-chaos-live/SUMMARY.txt
19		PASS	

#	Challenge	Verdict	Evidence
	HOTA-TELEMETRY-SCHEMA-LIVE		HOTA-TELEMETRY-SCHEMA-LIVE.log (12/0/0) + docs/qa/20260623-phase2-challenges/telemetry-schema/SUMMARY.txt
20	HOTA-ROLLOUT-STAGED-LIVE	PASS	HOTA-ROLLOUT-STAGED-LIVE.log (47/0/0) + docs/qa/20260623-phase2-challenges/rollout-staged/SUMMARY.txt

Tally

- **16 PASS** (vs live), **1 SKIP-with-reason** (off-topology, honest), **3 FAIL** (host QEMU+HVF accelerator condition — NOT helix-ota product defects).
- The 6 thinker live-system OTA challenges (TRUST-BOUNDARY, HTTP-LOAD, CHAOS, TELEMETRY-SCHEMA, ROLLOUT-STAGED + the F-CLUSTER-backed paths) all PASS vs live.

Honest findings (§11.4.6)

1. **AB-virt trio (12/13/14) FAIL — host QEMU+HVF accelerator non-functional.** The SAME unchanged A/B image booted fully on 2026-06-11 (committed 196-line console transcript). On this run QEMU+HVF emitted ZERO kernel output before the expect timeout; a standalone `qemu-system-aarch64 -accel hvf` sanity run also produced no kernel output in 20s. Host load avg ~9.5, 3-day uptime, 8 users. Root cause = HVF accelerator unavailable/contended on this Apple-Silicon host right now — NOT a helix-ota defect, NOT an A/B-image regression, NOT a test-script defect. Minor test-robustness gap: the SKIP guard checks qemu *presence* but not HVF *functionality*, so a dead accelerator hard-FAILs instead of exit-3 SKIPping. Image prereqs (`out/.ok`, images) WERE present.

2. **Boot-harness leftover-container race (intermittent, self-recovering).** `tests/lib/boot_real_system.sh` idempotent re-run hits `container state improper` / `name already in use` on a stale postgres container; it self-recovers and reaches `/readyz=200`. First trust-boundary in-challenge boot returned BASE_URL prematurely (~11s) so the challenge's own 6s readyz probe got 000 → honest SKIP `network_unreachable_external` (correct anti-bluff behavior, no fake-pass). Re-run detached with a pre-clean → PASS 4/0/0.

3. **Port collisions are harness artifacts, not defects.** PIPELINE-SIGNED and SECURITY-PROBES default to port 8080 when invoked directly; the sibling `htCore` owns mac :8080. The runner injects `HELIX_PORT=<free>` for self-hosting challenges, avoiding this. Re-run with a free port → both PASS.

Cleanup (§11.4.14, project-scoped)

- Shared :18091 ota-server killed (incl. lingering `go run` child); :18091 free.
- thinker F-CLUSTER project `helix-ota-system` torn down (0 containers).

- Sibling stacks UNTOUCHED: `lava-postgres-thinker` + `lava-api-go-thinker` (Up), mac `htCore` :8080 (Up). No stray ssh forwards.
- §11.4.10: no tokens/secrets in ledger (test scripts redact; bank-run admin password was env-only, never logged).